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865 (concluded). On Multiple Algebra.

(Continuation of §46; printed page 481.)

depending on the new extraordinaries e_1e_2, e_2e_3, e_3e_1 . Grassmann further assumes

$$e_1^2 = 0, \quad e_2^2 = 0, \quad e_3^2 = 0,$$

$$e_2e_3 = -e_3e_2 = \eta_1, \quad e_3e_1 = -e_1e_3 = \eta_2, \quad e_1e_2 = -e_2e_1 = \eta_3;$$

he calls the foregoing combinations the complements (*Ergänzungen*) of e_1, e_2, e_3 , denoting them by $/e_1, /e_2, /e_3$ respectively; it is better to use new letters, and I call them η_1, η_2, η_3 respectively; we thus have

$$e_2e_3 = -e_3e_2 = \eta_1, \quad e_3e_1 = -e_1e_3 = \eta_2, \quad e_1e_2 = -e_2e_1 = \eta_3.$$

But we require further assumptions for the combinations of the η 's *inter se* and with the e 's; the forms are

$$\begin{aligned} \eta_1e_1 &= \eta_2e_2 = \eta_3e_3 = 1, \\ -\eta_2\eta_3 &= \eta_3\eta_2 = e_1, \quad -\eta_3\eta_1 = \eta_1\eta_3 = e_2, \quad -\eta_1\eta_2 = \eta_2\eta_1 = e_3, \\ e_1\eta_1 &= \eta_1e_1 = 1, \quad e_2\eta_2 = \eta_2e_2 = 1, \quad e_3\eta_3 = \eta_3e_3 = 0, \quad \&c., \end{aligned}$$

(viz. each such combination is $= 0$); or, what is the same thing, we have for $1, e_1, e_2, e_3, \eta_1, \eta_2, \eta_3$ the multiplication table

	1	e_1	e_2	e_3	η_1	η_2	η_3
1	1	e_1	e_2	e_3	η_1	η_2	η_3
e_1	e_1	0	η_3	$-\eta_2$	1	0	0
e_2	e_2	$-\eta_3$	0	η_1	0	1	0
e_3	e_3	η_2	$-\eta_1$	0	0	0	1
η_1	η_1	1	0	0	0	$-e_3$	e_2
η_2	η_2	0	1	0	e_3	0	$-e_1$
η_3	η_3	0	0	1	$-e_2$	e_1	0

It is proper to remark that in Grassmann's theory there is no interpretation for the general linear symbol

$$\omega + x_1e_1 + x_2e_2 + x_3e_3 + y_1\eta_1 + y_2\eta_2 + y_3\eta_3,$$

which presents itself in connexion with the seven extraordinaries $(1, e_1, e_2, e_3, \eta_1, \eta_2, \eta_3)$.

47. It is to be noticed that the symbols are not associative; assuming them to be so, we should have $e_2e_1 \cdot e_2e_3 = e_2 \cdot e_1^2 \cdot e_3 = 0$, which is inconsistent with $\eta_3\eta_1 = -e_2$; and so in other cases. But any *three* e 's are associative, and hence a product

$$x \cdot y \cdot z = (x_1e_1 + x_2e_2 + x_3e_3)(y_1e_1 + y_2e_2 + y_3e_3)(z_1e_1 + z_2e_2 + z_3e_3),$$

is associative; in fact, if this be regarded as standing for $(x \cdot y)z$, the value is

$$= \{(x_2y_3 - x_3y_2)\eta_1 + (x_3y_1 - x_1y_3)\eta_2 + (x_1y_2 - x_2y_1)\eta_3\}(z_1e_1 + z_2e_2 + z_3e_3),$$

which is

$$= (x_2y_3 - x_3y_2)z_1 + (x_3y_1 - x_1y_3)z_2 + (x_1y_2 - x_2y_1)z_3 = \begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{vmatrix};$$

and similarly, regarding it as standing for $x(y \cdot z)$, the value is

$$= x_1(y_2z_3 - y_3z_2) + x_2(y_3z_1 - y_1z_3) + x_3(y_1z_2 - y_2z_1) = \begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{vmatrix},$$

as before. The product of the three points is thus a scalar, viz. it is equal to the area of the triangle formed by the three points divided by that of the triangle $A_1A_2A_3$.

48. But a product $x \cdot Y \cdot z$,

$$= (x_1e_1 + x_2e_2 + x_3e_3)(Y_1\eta_1 + Y_2\eta_2 + Y_3\eta_3)(z_1e_1 + z_2e_2 + z_3e_3),$$

is not associative, and has thus no meaning until the grouping of the factors is determined. In fact, $(x \cdot Y)z$ will be

$$= (x_1Y_1 + x_2Y_2 + x_3Y_3)(z_1e_1 + z_2e_2 + z_3e_3),$$

which denotes the point z_1 with a weight $= x_1Y_1 + x_2Y_2 + x_3Y_3$; whereas $x(Y \cdot z)$ will be

$$= (x_1e_1 + x_2e_2 + x_3e_3)(y_1Z_1 + y_2Z_2 + y_3Z_3),$$

which denotes the point x_1 with a weight $= y_1Z_1 + y_2Z_2 + y_3Z_3$.

Hence also a product $x \cdot y \cdot z \cdot w$,

$$= (x_1e_1 + x_2e_2 + x_3e_3)(y_1e_1 + y_2e_2 + y_3e_3)(z_1e_1 + z_2e_2 + z_3e_3)(w_1e_1 + w_2e_2 + w_3e_3),$$

of four factors is not associative, and has thus no meaning until the grouping of the factors is determined. Grassmann attributes to such a product the value $= 0$; but this is not a value corresponding to any grouping of the factors, and the equation $x \cdot y \cdot z \cdot w = 0$ can only be regarded as an independent assumption.

For the product of a point into itself, or say the square of a point, we have

$$x^2 = (x_1e_1 + x_2e_2 + x_3e_3)^2 = (x_1e_1 + x_2e_2 + x_3e_3)(x_1e_1 + x_2e_2 + x_3e_3) = 0,$$

viz. this is $=$ zero.

49. The product of two points x and y is defined to be the “Strecke,” the finite line or stroke (xy) , considered as drawn from x to y , say we have $x \cdot y = (xy)$; observe

that $y \cdot x = (yx)$: by what precedes $y \cdot x + x \cdot y = 0$, whence also $(yx) + (xy) = 0$, which is an equation in the addition of strokes. Hence, if as before, the points x, y are $x = x_1e_1 + x_2e_2 + x_3e_3$ and $y = y_1e_1 + y_2e_2 + y_3e_3$, then for the stroke (xy) we have

$$\begin{aligned}(xy) &= (x_1e_1 + x_2e_2 + x_3e_3)(y_1e_1 + y_2e_2 + y_3e_3), \\ &= (x_2y_3 - x_3y_2)\eta_1 + (x_3y_1 - x_1y_3)\eta_2 + (x_1y_2 - x_2y_1)\eta_3,\end{aligned}$$

which is, in fact,

$$= (A_1xy \cdot A_1 + A_2xy \cdot A_2 + A_3xy \cdot A_3) \div A_1A_2A_3.$$

if $A_1xy, A_2xy, A_3xy, A_1A_2A_3$ denote the areas of the triangles formed by the points $(A_1, x, y), (A_2, x, y), (A_3, x, y), (A_1, A_2, A_3)$ respectively. These triangles remain unaltered if the stroke (xy) is slid along the indefinite line joining the original points x, y , the absolute distance xy being unaltered; that is, strokes of equal length upon the same line are regarded as identical. Hence also strokes on the same line may be added together by adding their lengths.

50. Consider two strokes $(xy), (xz)$, (fig. 4), having a common point x , then completing the parallelogram, we prove that the sum $(xy) + (xz)$ is equal to the diagonal (xw) .

[Figure 4: a parallelogram with vertex x at lower-left, y at lower-right, z at upper-left, w at upper-right; both diagonals drawn, illustrating $x + w = y + z$.]

We have, in fact,

$$x + w = y + z,$$

and thence

$$x(x + w) = x(y + z),$$

that is,

$$x^2 + x \cdot w = x \cdot y + x \cdot z,$$

or, since $x^2 = 0$ and $x \cdot y = (xy)$, &c., this is

$$(xw) = (xy) + (xz),$$

the required formula for the addition of the strokes.

51. The same thing appears thus: writing

$$\begin{aligned}x &= x_1e_1 + x_2e_2 + x_3e_3, \\ y &= y_1e_1 + y_2e_2 + y_3e_3, \\ z &= z_1e_1 + z_2e_2 + z_3e_3,\end{aligned}$$

we have

$$\begin{aligned}(xy) &= (x_2y_3 - x_3y_2)\eta_1 + (x_3y_1 - x_1y_3)\eta_2 + (x_1y_2 - x_2y_1)\eta_3, \\ (xz) &= (x_2z_3 - x_3z_2)\eta_1 + (x_3z_1 - x_1z_3)\eta_2 + (x_1z_2 - x_2z_1)\eta_3.\end{aligned}$$

Or, putting for a moment

$$m_1 = \frac{1}{2}(y_1 + z_1), \quad m_2 = \frac{1}{2}(y_2 + z_2), \quad m_3 = \frac{1}{2}(y_3 + z_3),$$

we have for the point m , midway between y and z , the equation $m = m_1e_1 + m_2e_2 + m_3e_3$; and then

$$\begin{aligned} (xy) + (xz) &= 2\{(x_2m_3 - x_3m_2)\eta_1 + (x_3m_1 - x_1m_3)\eta_2 + (x_1m_2 - x_2m_1)\eta_3\} \\ &= 2(xm), = xw, \end{aligned}$$

which is the theorem in question.

52. Recurring to the foregoing equation $x + w = y + z$, we deduce

$$xz - yw = zw - xy = (yzw) = (zwx) = (wxy) = (xyz),$$

if for shortness $(yzw) = yz + zw + wy$, and similarly for (zwx) , (wxy) , and (xyz) . Observe here that $xz - yw$ and $zw - xy$ are each of them equal to the difference between two equal and parallel strokes; the strokes yz , zw , wy are the sides (in order) of a triangle; qua forces, the expressions in question represent each of them a couple.

53. We have thus arrived at the representation of a stroke

$$(xy) = (x_2y_3 - x_3y_2)\eta_1 + (x_3y_1 - x_1y_3)\eta_2 + (x_1y_2 - x_2y_1)\eta_3,$$

or say

$$\lambda = \lambda_1\eta_1 + \lambda_2\eta_2 + \lambda_3\eta_3,$$

where the meaning of the coordinates λ_1 , λ_2 , λ_3 is such that, if x , y are the two extremities of the stroke, then λ_1 , λ_2 , λ_3 are equal to the areas A_1xy , A_2xy , A_3xy , each divided by the area $A_1A_2A_3$. And we have also established the rule for the addition of strokes: it will be noticed that this has been done without the aid of any expression for the length of the stroke.

54. *Metrical relations.* For the expression of the distance of two points or length of a stroke, and of the inclinations of a stroke to the sides of the fundamental triangle, &c., we require the values of the angles and sides of this triangle $A_1A_2A_3$, say the angles are A_1 , A_2 , A_3 , and the sides are $= R \sin A_1$, $R \sin A_2$, $R \sin A_3$ respectively; R is thus equal to the diameter of the circumscribed circle. Moreover, considering a stroke xy , if through x we draw lines $x\theta_1$, $x\theta_2$, $x\theta_3$ in the senses A_1A_1 , A_1A_2 , A_1A_3 respectively, then taking the angles $yx\theta_1$, $yx\theta_2$, $yx\theta_3$, each of them in the same sense, or say each right-handedly, we call these angles θ_1 , θ_2 , θ_3 respectively, and take also ρ , $= (xy)$ for the length of the stroke. This being so, and if the two extremities are

$$x = x_1e_1 + x_2e_2 + x_3e_3, \quad \text{and} \quad y = y_1e_1 + y_2e_2 + y_3e_3,$$

then we have, first,

$$\theta_1 - \theta_2 = \pi - A_3, \quad \theta_2 - \theta_3 = \pi - A_1, \quad \theta_3 - \theta_1 = \pi - A_2,$$

giving

$$A_1 + A_2 + A_3 = \pi,$$

and

$$\sin A_1 = \sin(\theta_2 - \theta_3), \quad \sin A_2 = \sin(\theta_3 - \theta_1), \quad \sin A_3 = \sin(\theta_1 - \theta_2).$$

And then

$$y_1 = x_1 + \frac{\rho}{R} \frac{\sin \theta_1}{\sin A_2 \sin A_3}, \quad y_2 = x_2 + \frac{\rho}{R} \frac{\sin \theta_2}{\sin A_3 \sin A_1}, \quad y_3 = x_3 + \frac{\rho}{R} \frac{\sin \theta_3}{\sin A_1 \sin A_2}.$$

As regards these last equations, writing for a moment p_1, p_2, p_3 for the perpendicular distances of x , and q_1, q_2, q_3 for those of y from the sides of the triangle, then we have $q_1, q_2, q_3 = p_1 + \rho \sin \theta_1, p_2 + \rho \sin \theta_2, p_3 + \rho \sin \theta_3$, and we thence easily derive the equations in question.

55. *Expression for length of a stroke.* We hence, by a somewhat complicated analytical process, find an expression for ρ in terms of the coordinates (x_1, x_2, x_3) and (y_1, y_2, y_3) , which enter into it through the combinations

$$x_2 y_3 - x_3 y_2, \quad x_3 y_1 - x_1 y_3, \quad x_1 y_2 - x_2 y_1.$$

In fact, writing for shortness

$$\frac{x_1}{\sin A_1}, \quad \frac{x_2}{\sin A_2}, \quad \frac{x_3}{\sin A_3} = K_1, \quad K_2, \quad K_3,$$

we have

$$\begin{aligned} (x_2 y_3 - x_3 y_2) \sin A_1 &= \frac{\rho}{R} (K_2 \sin \theta_3 - K_3 \sin \theta_2), \\ (x_3 y_1 - x_1 y_3) \sin A_2 &= \frac{\rho}{R} (K_3 \sin \theta_1 - K_1 \sin \theta_3), \\ (x_1 y_2 - x_2 y_1) \sin A_3 &= \frac{\rho}{R} (K_1 \sin \theta_2 - K_2 \sin \theta_1); \end{aligned}$$

and hence also, if $(* | \xi, \eta, \zeta)^2$ denote the quadric function

$$(1, 1, 1, -\cos A_1, -\cos A_2, -\cos A_3 | \xi, \eta, \zeta)^2,$$

we have

$$(* | (x_2 y_3 - x_3 y_2) \sin A_1, \dots)^2 = \frac{\rho^2}{R^2} (* | K_2 \sin \theta_3 - K_3 \sin \theta_2, K_3 \sin \theta_1 - K_1 \sin \theta_3, K_1 \sin \theta_2 - K_2 \sin \theta_1)^2,$$

where the quadric function on the right-hand side is in fact

$$= (K_1 \sin A_1 + K_2 \sin A_2 + K_3 \sin A_3)^2,$$

that is,

$$= (x_1 + x_2 + x_3)^2, = 1;$$

and we thus finally obtain

$$\rho^2 = R^2 (* | (x_2 y_3 - x_3 y_2) \sin A_1, (x_3 y_1 - x_1 y_3) \sin A_2, (x_1 y_2 - x_2 y_1) \sin A_3)^2,$$

the required formula for the length of the stroke xy .

56. To effect the foregoing reduction of the quadric function

$$(* | K_2 \sin \theta_3 - K_3 \sin \theta_2, K_3 \sin \theta_1 - K_1 \sin \theta_3, K_1 \sin \theta_2 - K_2 \sin \theta_1)^2,$$

observe that the coefficient herein of K_1^2 is

$$= \sin^2 \theta_2 + \sin^2 \theta_3 + 2 \cos A_1 \sin \theta_2 \sin \theta_3,$$

where $\theta_2 - \theta_3 = \pi - A_1$, and thence

$$\cos \theta_2 \cos \theta_3 + \sin \theta_2 \sin \theta_3 = -\cos A_1;$$

that is, we have

$$\cos A_1 + \sin \theta_2 \sin \theta_3 = -\cos \theta_2 \cos \theta_3,$$

and thence

$$1 - \sin^2 \theta_2 - \sin^2 \theta_3 + \sin^2 \theta_2 \sin^2 \theta_3 = \cos^2 A_1 + 2 \cos A_1 \sin \theta_2 \sin \theta_3 + \sin^2 \theta_2 \sin^2 \theta_3;$$

whence the coefficient in question is $= \sin^2 A_1$. And finding in like manner the coefficients of K_2^2 , K_3^2 , $K_2 K_3$, &c., the whole function is, as already mentioned,

$$= (K_1 \sin A_1 + K_2 \sin A_2 + K_3 \sin A_3)^2, \text{ that is, } = 1.$$

57. *New representation of a stroke.* Representing the stroke in the form

$$\lambda = \lambda_1 \eta_1 + \lambda_2 \eta_2 + \lambda_3 \eta_3,$$

we have only to replace $x_2 y_3 - x_3 y_2$, $x_3 y_1 - x_1 y_3$, $x_1 y_2 - x_2 y_1$ by their values $= \lambda_1, \lambda_2, \lambda_3$ respectively, viz. for the length ρ of the stroke, we have

$$\rho^2 = R^2 (* | \lambda_1 \sin A_1, \lambda_2 \sin A_2, \lambda_3 \sin A_3)^2.$$

Say this equation is

$$\rho^2 = R^2 \cdot \Omega, \text{ then } \rho = R \sqrt{(\Omega)}, \text{ or } 1 = R \frac{\rho}{\sqrt{(\Omega)}},$$

and the stroke may be represented in the form

$$\lambda = \frac{\rho}{R \sqrt{(\Omega)}} (\lambda_1 \eta_1 + \lambda_2 \eta_2 + \lambda_3 \eta_3);$$

or, what is the same thing, if the absolute magnitudes of the coordinates $\lambda_1, \lambda_2, \lambda_3$ are such that

$$(* | \lambda_1 \sin A_1, \lambda_2 \sin A_2, \lambda_3 \sin A_3)^2 = 1,$$

then the expression is

$$\lambda = \frac{\rho}{R} (\lambda_1 \eta_1 + \lambda_2 \eta_2 + \lambda_3 \eta_3),$$

where ρ is the length of the stroke.

58. It has been seen that the coefficients of η_1, η_2, η_3 have the values $A_1 x y, A_2 x y, A_3 x y$, each divided by $A_1 A_2 A_3$; or, what is the same thing, if p_1, p_2, p_3 are the lengths of the perpendiculars on the stroke from the points A_1, A_2, A_3 respectively, then these coefficients are $(\frac{1}{2} \rho p_1, \frac{1}{2} \rho p_2, \frac{1}{2} \rho p_3)$, each divided by $A_1 A_2 A_3$; or, what is the same thing, the perpendicular distances p_1, p_2, p_3 are

$$= \frac{2 A_1 A_2 A_3}{R \sqrt{(\Omega)}} (\lambda_1, \lambda_2, \lambda_3)$$

respectively, where $A_1 A_2 A_3$ is the area of the fundamental triangle.

59. *Expression for the mutual inclination of two strokes.* In connexion with the stroke $\lambda = \lambda_1\eta_1 + \lambda_2\eta_2 + \lambda_3\eta_3$, considering a stroke $\mu = \mu_1\eta_1 + \mu_2\eta_2 + \mu_3\eta_3$, suppose the inclinations hereof to the axes are ϕ_1, ϕ_2, ϕ_3 , then

$$\phi_1 - \phi_2 = \pi - A_3, \quad \phi_2 - \phi_3 = \pi - A_1, \quad \phi_3 - \phi_1 = \pi - A_2,$$

and we have

$$\theta_1 - \phi_1 = \theta_2 - \phi_2 = \theta_3 - \phi_3,$$

viz. each of these equal angles is the inclination of the two strokes to each other, say this angle is $= \delta$. The expression for this angle is

$$\cos \delta = (* | \lambda_1 \sin A_1, \lambda_2 \sin A_2, \lambda_3 \sin A_3 | \mu_1 \sin A_1, \mu_2 \sin A_2, \mu_3 \sin A_3) \\ \div \sqrt{(1, \dots | \lambda_1 \sin A_1, \lambda_2 \sin A_2, \lambda_3 \sin A_3)^2} \cdot \sqrt{(1, \dots | \mu_1 \sin A_1, \mu_2 \sin A_2, \mu_3 \sin A_3)^2}.$$

In fact, considering (as we may do) the two strokes as proceeding from a common point $x = x_1e_1 + x_2e_2 + x_3e_3$, then the function in question is

$$= (* | K_2 \sin \theta_3 - K_3 \sin \theta_2, K_3 \sin \theta_1 - K_1 \sin \theta_3, K_1 \sin \theta_2 - K_2 \sin \theta_1 | K_2 \sin \phi_3 - K_3 \sin \phi_2, \dots),$$

which is

$$= (K_1 \sin A_1 + K_2 \sin A_2 + K_3 \sin A_3)^2 \cos \delta, = \cos \delta.$$

60. In verification, observe that the whole coefficient of K_1^2 is

$$= \sin \theta_2 \sin \phi_3 + \sin \theta_3 \sin \phi_2 + \cos A_1 (\sin \theta_2 \sin \phi_3 + \sin \theta_3 \sin \phi_2),$$

where $\theta_2 - \theta_3 = \pi - A_1, \phi_2 - \phi_3 = \pi - A_1$. Hence

$$\sin \theta_3 = -\sin(\theta_2 - A_1), \quad \sin \phi_3 = -\sin(\phi_2 - A_1);$$

and the expression in question is

$$\begin{aligned} &= \sin(\theta_2 - A_1) \sin(\phi_2 - A_1) \\ &\quad - \cos A_1 \sin(\theta_2 - A_1) \sin \phi_2 \\ &\quad - \cos A_1 \sin(\phi_2 - A_1) \sin \theta_2 \\ &\quad + \sin \theta_2 \sin \phi_2, \end{aligned}$$

which, expanding the sines, and writing the last term in the form

$$= \sin \theta_2 \sin \phi_2 (\cos^2 A_1 + \sin^2 A_1)$$

is

$$\begin{aligned} &= \cos^2 A_1 (\sin \theta_2 \sin \phi_2 - \sin \theta_2 \sin \phi_2 - \sin \theta_2 \sin \phi_2 + \sin \theta_2 \sin \phi_2) \\ &\quad + \sin^2 A_1 (\cos \theta_2 \cos \phi_2 + \sin \theta_2 \sin \phi_2) \\ &\quad + \cos A_1 \sin A_1 (-\sin \theta_2 \cos \phi_2 - \sin \phi_2 \cos \theta_2 + \sin \phi_2 \cos \theta_2 + \sin \theta_2 \cos \phi_2), \\ &= \sin^2 A_1 \cos(\theta_2 - \phi_2), = \sin^2 A_1 \cos \delta, \end{aligned}$$

and reducing in like manner the coefficients of K_2^2, K_3^2, K_2K_3 , &c., the whole expression becomes

$$= (K_1 \sin A_1 + K_2 \sin A_2 + K_3 \sin A_3)^2 \cos \delta, = \cos \delta \text{ as above.}$$

61. The foregoing formulae agree with the theorem, No. 50, that the sum of the strokes

$$\lambda_1 = \frac{\rho}{R\sqrt{(\Omega)}} (\lambda_1\eta_1 + \lambda_2\eta_2 + \lambda_3\eta_3),$$

and

$$\mu_1 = \frac{\sigma}{R\sqrt{(\Upsilon)}} (\mu_1\eta_1 + \mu_2\eta_2 + \mu_3\eta_3),$$

is a stroke

$$= \frac{\tau}{R\sqrt{(\Pi)}} (\nu_1\eta_1 + \nu_2\eta_2 + \nu_3\eta_3),$$

which is the diagonal of the parallelogram constructed with the given strokes λ and μ ; the proof is a little simplified by assuming (as is allowable) $\Omega = 1$ and $\Upsilon = 1$, in which case we have

$$\lambda + \mu = \frac{1}{R} [(\rho\lambda_1 + \sigma\mu_1)\eta_1 + (\rho\lambda_2 + \sigma\mu_2)\eta_2 + (\rho\lambda_3 + \sigma\mu_3)\eta_3],$$

from which the length and inclination may be calculated.

As already appearing, the product x, y, z of any three points is the scalar

$$\begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{vmatrix},$$

which is equal to the area of the triangle xyz , divided by that of the triangle $A_1A_2A_3$. We have in like manner the product of a point x into a stroke λ , viz. this is

$$= (x_1\lambda_1 + x_2\lambda_2 + x_3\lambda_3)(\lambda_1\eta_1 + \lambda_2\eta_2 + \lambda_3\eta_3),$$

which is $= x_1\lambda_1 + x_2\lambda_2 + x_3\lambda_3$, the two factors being in this case commutative; the value is equal to the area of the triangle formed by the point and the stroke, divided by the area $A_1A_2A_3$. Of course, if in the one case the three points are in a line, or if in the other the point is in the line of the stroke, then the product is $= 0$.

62. We have yet to consider the product of two strokes; say these are

$$\lambda = \lambda_1\eta_1 + \lambda_2\eta_2 + \lambda_3\eta_3, \quad \text{and} \quad \mu = \mu_1\eta_1 + \mu_2\eta_2 + \mu_3\eta_3,$$

then we have

$$\begin{aligned} \lambda \cdot \mu &= (\lambda_1\eta_1 + \lambda_2\eta_2 + \lambda_3\eta_3)(\mu_1\eta_1 + \mu_2\eta_2 + \mu_3\eta_3) \\ &= -[(\lambda_2\mu_3 - \lambda_3\mu_2)e_1 + (\lambda_3\mu_1 - \lambda_1\mu_3)e_2 + (\lambda_1\mu_2 - \lambda_2\mu_1)e_3], \end{aligned}$$

which is a point regarded as having weight. If we take the two strokes to be

$$\begin{aligned} \lambda &= (xy) = (x_2y_3 - x_3y_2)\eta_1 + (x_3y_1 - x_1y_3)\eta_2 + (x_1y_2 - x_2y_1)\eta_3, \\ \text{and } \mu &= (xz) = (x_2z_3 - x_3z_2)\eta_1 + (x_3z_1 - x_1z_3)\eta_2 + (x_1z_2 - x_2z_1)\eta_3, \end{aligned}$$

then the product is

$$\lambda \cdot \mu = -(x_1 e_1 + x_2 e_2 + x_3 e_3) \begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{vmatrix},$$

viz. this is the common point x , with a weight = area of triangle $xyz \div$ area $A_1 A_2 A_3$, or say = the area of triangle formed by the two strokes $\div$ area $A_1 A_2 A_3$. The product is non-commutative: we, in fact, have $\lambda \cdot \mu = -\mu \cdot \lambda$.

63. The product of three strokes is associative; and it is a scalar. If the three strokes are the sides of a triangle LMN , then the value is $= (LMN)^2 \div (A_1 A_2 A_3)^3$, where LMN and $A_1 A_2 A_3$ denote the areas of the triangle LMN and the triangle $A_1 A_2 A_3$ respectively; and if any stroke instead of being a side of the triangle LMN be a part only of this side, then the value is diminished proportionally; hence the value is $= \frac{\rho\sigma\tau}{MN \cdot NL \cdot LM} \{(LMN)^2 \div (A_1 A_2 A_3)^3\}$, where ρ, σ, τ are the lengths of the three strokes, MN, NL, LM the lengths of the sides of the triangle LMN , $A_1 A_2 A_3$ are the areas as above.

64. It is to be remarked that the form

$$(* | \xi, \eta, \zeta)^2, = (1, 1, 1, -\cos A_1, -\cos A_2, -\cos A_3 | \xi, \eta, \zeta)^2$$

is the product of two linear factors; it is

$$= (\xi - \eta \cos A_3 - \zeta \cos A_2)^2 + (\eta \sin A_3 - \zeta \sin A_2)^2,$$

and thus

$$= (\xi - \eta e^{iA_3} - \zeta e^{-iA_2})(\xi - \eta e^{-iA_3} - \zeta e^{iA_2}).$$

This corresponds to the theorem that the distance of two points is $= 0$, when the line joining them passes through one of the circular points at infinity; the coordinates of one of these points may be written under the three equivalent forms

$$\begin{aligned} x_1 : x_2 : x_3 &= \sin A_1 : -\sin A_2 e^{-iA_3} : -\sin A_3 e^{-iA_2}, \\ &= -\sin A_1 e^{-iA_3} : \sin A_2 : -\sin A_3 e^{iA_1}, \\ &= -\sin A_1 e^{iA_2} : -\sin A_2 e^{iA_1} : \sin A_3, \end{aligned}$$

and those of the other are of course obtained therefrom by the change of i into $-i$.

*(To be continued)**

[* This paper remains incomplete; no continuation appears to have been prepared by Professor Cayley.]

866. Note on Kiepert's L -Equations, in the Transformation of Elliptic Functions.

[From the *Mathematische Annalen*, t. xxx. (1887), pp. 75–77.]

IT APPEARS, by comparison with Klein's paper "Ueber die Transformation u. s. w.," *Math. Annalen*, t. xiv. (1878), see p. 144, that Kiepert's L made use of in the Memoir "Ueber Theilung und Transformation der elliptischen Functionen," *Math. Annalen*, t. xxvi. (1886), pp. 369–454, is, in fact, the square of the multiplier, "für das durch $\sqrt[5]{\Delta}$ normirte Integral," viz. considering the general quartic function $(a, \dots)(x, 1)^4 = (a, b, c, d, e)(x, 1)^4$ and the transformed function $(a_1, \dots)(y, 1)^4$, then we have

$$\frac{L \sqrt[5]{\Delta} dx}{\sqrt{(a, \dots)(x, 1)^4}} = \frac{\sqrt[5]{\Delta_1} dy}{\sqrt{(a_1, \dots)(y, 1)^4}},$$

where if

$$\begin{aligned} I &= ae - 4bd + 3c^2, \\ J &= ace - ad^2 - b^2e + 2bcd - c^3, \end{aligned}$$

and similarly I_1, J_1 , are the invariants of the two functions, then Δ, Δ_1 , are the discriminants

$$\Delta = I^3 - 27J^2, \quad \Delta_1 = I_1^3 - 27J_1^2,$$

and the γ_1, γ_1 of Kiepert's equations are

$$\gamma_1 = I + \sqrt[3]{\Delta}, \quad \gamma_1 = J + \sqrt{\Delta},$$

whence

$$\gamma_1^3 - 27\gamma_1^2 = 1.$$

In particular, if the forms are

$$1 - x^5, \quad 1 - k^2 x^6, \quad \text{and} \quad 1 - y^5 \cdot 1 - \lambda^2 y^6,$$

and if as usual $k = u^4$, $\lambda = v^4$, and M is the multiplier for the form

$$\frac{dx}{\sqrt{1-x^2 \cdot 1 - k^2 x^2}} = \frac{M dy}{\sqrt{1-y^2 \cdot 1 - \lambda^2 y^2}},$$

then we have

$$\begin{aligned} I &= \frac{1}{12} (1 + 14u^8 + u^{10}), \\ J &= \frac{1}{216} (1 + u^8) (1 - 34u^8 + u^{16}), \\ \Delta &= \frac{1}{16} u^8 (1 - u^8)^4, \quad \Delta_1 = \frac{1}{16} v^8 (1 - v^8)^4, \\ \gamma_1 &= \frac{1}{4} \sqrt[5]{2} \frac{1 + 14u^8 + u^{16}}{u^4 (1 - u^8)^4}, \quad \gamma_1 = \frac{1}{4} \frac{(1 + u^8) (1 - 34u^8 + u^{16})}{u^4 (1 - u^8)^2}. \end{aligned}$$

and thence

$$L^2 = \frac{v^8 (1 - v^8)^{\frac{4}{5}}}{u^8 (1 - u^8)^{\frac{4}{5}}} \frac{1}{M},$$

which last equation is the expression for L^2 in terms of the Jacobian symbols u , v , M .

As an easy verification in a particular case, suppose $n = 5$. We have here

$$L^2 = \frac{v^8 (1 - v^8)^{\frac{4}{5}}}{u^8 (1 - u^8)^{\frac{4}{5}}} \cdot \frac{1}{M}, \quad M = \frac{v (1 - uv^6)}{v - u^5}, \quad \left(= \frac{v + u^5}{5u (1 + u^8 v^8)} \right),$$

$$u^2 - v^2 + 5u^2 v^2 (u^2 - v^2) + 4uv (1 - u^4 v^4) = 0,$$

$$\gamma_1 = \frac{1}{4} \sqrt[5]{2} \frac{1 + 14u^8 + u^{16}}{u^4 (1 - u^8)^4};$$

and it should be possible, by eliminating u , v , M , to deduce hence the L -equation

$$L^6 + 10L^5 - 12\gamma_1 L^2 + 5 = 0. \quad (\text{Kiepert, p. 428.})$$

It does not seem in any wise easy to do this in the case of an arbitrary modulus; but writing the modular equation in the form

$$(u^6 - v^6)(u^6 + 6u^2 v^2 + v^2) + 4uv (1 - u^4 v^4) = 0,$$

we satisfy this by

$$uv - 1 = 0, \quad u^6 + 6u^2 v^2 + v^2 = 0,$$

or say by

$$v = \frac{1}{u}, \quad u^8 + 6u^4 + 1 = 0,$$

and the equation may be verified for this particular modulus.

We have

$$1 + 14u^8 + u^{16} = 48u^8, \quad (1 - u^8)^4 = 32u^8,$$

and consequently

$$\gamma_1 = \frac{1}{4} \sqrt[5]{2} \cdot \frac{48u^8}{u^4 (32u^8)^4} = \frac{1}{2 \cdot 3} \cdot 2^{\frac{6}{5}} \frac{2^4 \cdot 3u^8}{u^4 \cdot 2^{\frac{12}{5}} \cdot u^{32}} = 1, \quad (\text{whence also } \gamma_1 = 0).$$

Moreover

$$M = \frac{\frac{1}{u} - \frac{u}{u^4}}{\frac{1}{u} - u^5} = -\frac{1}{u^2} \frac{1 - u^2}{1 - u^6} = \frac{-1}{u^2 (1 + u^2 + u^4)},$$

and thence

$$L^2 = \frac{\frac{1}{u^8} \left(1 - \frac{1}{u^8}\right)^{\frac{4}{5}}}{u^8 (1 - u^8)^{\frac{4}{5}}} \cdot \frac{1}{M} = \frac{(u^8 - 1)^{\frac{4}{5}}}{u^4 (1 - u^8)^{\frac{4}{5}}} \cdot \frac{1}{M} = -\frac{1}{u^2 M},$$

that is,

$$L^2 = 1 + u^2 + u^{-2}.$$

But we have

$$u^4 = -3 + 2\sqrt{2},$$

and thence

$$u^2 = i(1 - \sqrt{2}), \quad u^{-2} = i(1 + \sqrt{2}),$$

and

$$L^2 = 1 + 2i,$$

whence

$$L^{10} + 10L^5 - 12\gamma_1 L^2 + 5 = (117 + 44i) + 10(-11 - 2i) - 12(1 + 2i) + 5 = 0,$$

or the L -equation is satisfied.

Cambridge, 14 March 1887.

867. Note on the Jacobian Sextic Equation.

[From the *Mathematische Annalen*, t. xxx. (1887), pp. 78–84.]

In the Jacobian sextic equation

$$(z-a)^6 - 4a(z-a)^5 + 10\delta(z-a)^2 - 4\epsilon(z-a) + 5\delta^2 - 4a\epsilon = 0,$$

if A, B, C, D, E, F are the square roots (each with a determinate sign) of the roots $z_1, z_2, z_3, z_4, z_5, z_6$ of the equation, and if ϵ be an imaginary fifth root of unity, such that $\epsilon + \epsilon^2 - \epsilon^3 - \epsilon^4 = \sqrt{5}$, then we have between the square roots a system of three linear equations

$$\begin{aligned} 0 &= -A\sqrt{5} + B + C + D + E + F, \\ 0 &= B + \epsilon C + \epsilon D + \epsilon^2 E + \epsilon^2 F, \\ 0 &= B + \epsilon^2 C + \epsilon^2 D + \epsilon E + \epsilon^2 F, \end{aligned}$$

see Briochi's *Funzioni ellittiche* (Milan, 1880), third appendix, p. 402.

The sextic equation is irreducible, and there is thus nothing to distinguish any one square root from any other square root. But in the foregoing system of equations, A is distinguished from the other square roots; hence the system must be one of 6 systems, wherein the place occupied by A is occupied by A, B, C, D, E, F respectively. Observe however that the letters being square roots, it may very well happen, and (as will appear) it does in fact happen, that the passage from the first to another system implies a change of sign in certain of the letters A, B, C, D, E, F .

The selection of one of the square roots as $= A$, does not determine which of the other square roots shall be $= B, C, D, E, F$ respectively: and in fact each system of equations may be represented in 10 different forms; viz. by multiplying the second and third equations only by $\epsilon, \epsilon^2, \epsilon^3, \epsilon^4$ respectively, we obtain four new forms: we have thus five forms: and then in each of them transposing the second and third equations, we have five other forms: we have thus in all 10 forms.

Write $ABCDEF$ to denote the foregoing system of three equations

$$\begin{aligned} 0 &= -A\sqrt{5} + B + C + D + E + F, \\ 0 &= B + \epsilon C + \epsilon D + \epsilon^2 E + \epsilon^2 F, \\ 0 &= B + \epsilon^2 C + \epsilon^2 D + \epsilon E + \epsilon^2 F; \end{aligned}$$

the 10 forms of the system are

$ABCDEF$	$ABFEDC$
$ACDEFB$	$ACBFED$
$ADEFBC$	$ADCBF E$
$AEFBCD$	$AEDCBF$
$AFBCDE$	$AFEDCB$,

where observe that the five forms in the same column are connected by cyclical substitutions on the last five letters; and that the forms in each line are connected by two inversions of the last four letters.

It is hardly necessary to remark that $ABFEDC$ means the system of three equations

$$\begin{aligned} 0 &= -A\sqrt{5} + B + F + E + D + C, \\ 0 &= B + \epsilon F + \epsilon E + \epsilon D + \epsilon C, \\ 0 &= B + \epsilon^2 F + \epsilon^2 E + \epsilon D + \epsilon C, \end{aligned}$$

which are the same with the equations of the original form; and similarly that $ACDEFB$ means the system of three equations

$$\begin{aligned} 0 &= -A\sqrt{5} + C + D + E + F + B, \\ 0 &= C + \epsilon D + \epsilon E + \epsilon^2 F + \epsilon^2 B, \\ 0 &= C + \epsilon^2 D + \epsilon^2 E + \epsilon F + \epsilon^2 B, \end{aligned}$$

where the first equation is the same with the first equation, and the second and third equations only differ by a factor from the second and third equations respectively of the original form. And similarly as regards the others of the 10 forms.

I say that we have a second system $BCAF\bar{D}\bar{E}$ (where $\bar{D}$, $\bar{E}$ mean $-D$, $-E$ respectively): in fact, this denotes the system of three equations

$$\begin{aligned} 0 &= -B\sqrt{5} + C + A + F - D - E, \\ 0 &= C + \epsilon A + \epsilon F - \epsilon^2 D - \epsilon^2 E, \\ 0 &= C + \epsilon^2 A + \epsilon^2 F - \epsilon D - \epsilon^2 E, \end{aligned}$$

which are deducible from the original three equations.

We have, in fact, the identities

$$\begin{aligned}
 -\sqrt{5}[-B\sqrt{5} + C + A + F - D - E] &= 1(B + C - A\sqrt{5} + F + D + E) \\
 &\quad + 2(B + \epsilon C + \epsilon F + \epsilon^2 D + \epsilon^2 E) \\
 &\quad + 2(B + \epsilon^2 C + \epsilon^2 F + \epsilon D + \epsilon E), \\
 (1 - \epsilon - \epsilon^2 + \epsilon^4)[C + \epsilon A + \epsilon F - \epsilon^2 D - \epsilon^2 E] \\
 &= 1(B + C - A\sqrt{5} + F + D + E) \\
 &\quad + (\epsilon + \epsilon^4)(B + \epsilon C + \epsilon F + \epsilon^2 D + \epsilon^2 E) \\
 &\quad + (\epsilon^2 + \epsilon^3)(B + \epsilon^2 C + \epsilon^2 F + \epsilon D + \epsilon E),
 \end{aligned}$$

and

$$\begin{aligned}
 (1 + \epsilon - \epsilon^2 - \epsilon^4)[C + \epsilon^2 A + \epsilon^2 F - \epsilon D - \epsilon E] \\
 &= 1(B + C - A\sqrt{5} + F + D + E) \\
 &\quad + (\epsilon^2 + \epsilon^3)(B + \epsilon C + \epsilon F + \epsilon^2 D + \epsilon^2 E) \\
 &\quad + (\epsilon + \epsilon^4)(B + \epsilon^2 C + \epsilon^2 F + \epsilon D + \epsilon E),
 \end{aligned}$$

all which identities are at once verified on recollecting that

$$1 = -\epsilon - \epsilon^2 - \epsilon^3 - \epsilon^4 \quad \text{and} \quad \sqrt{5} = \epsilon - \epsilon^2 - \epsilon^3 + \epsilon^4.$$

We can now write down the 6 systems, each of them under its 10 forms; these, in fact, are

$ABCDEF$	$ABFEDC$
$ACDEFB$	$ACBFED$
$ADEFBC$	$ADCBFE$
$AEFBCD$	$AEDCBF$
$AFBCDE$	$AFEDCB$
$BCAF\bar{D}\bar{E}$	$BC\bar{E}\bar{D}FA$
$BAF\bar{D}\bar{E}C$	$BAC\bar{E}\bar{D}F$
$BF\bar{D}\bar{E}CA$	$BFAC\bar{E}\bar{D}$
$B\bar{D}\bar{E}CAF$	$B\bar{D}FAC\bar{E}$
$B\bar{E}CAF\bar{D}$	$B\bar{E}\bar{D}FAC$
$CDABF\bar{E}$	$CDF\bar{E}BA$
$CABF\bar{E}D$	$CAD\bar{F}EB$
$CBF\bar{E}DA$	$CB\bar{A}D\bar{P}\bar{E}$
$C\bar{E}\bar{F}DAB$	$C\bar{E}DABF$
$CFDAB\bar{E}$	$CFEBAD$

(continuation of the table of 6 systems under 10 forms each:)

$DEAC\bar{F}\bar{B}$	$DE\bar{B}\bar{F}CA$
$DAC\bar{F}\bar{B}E$	$DCF\bar{B}EA$
$DC\bar{F}\bar{B}EA$	$DCAE\bar{B}\bar{F}$
$D\bar{F}\bar{B}EAC$	$D\bar{F}CAE\bar{B}$
$D\bar{B}EAC\bar{F}$	$D\bar{B}\bar{F}CAE$
$EFADB\bar{C}$	$EF\bar{C}BDA$
$EAD\bar{B}\bar{C}F$	$EAF\bar{C}\bar{B}D$
$ED\bar{B}\bar{C}FA$	$ED\bar{A}\bar{F}\bar{C}\bar{B}$
$E\bar{B}\bar{C}FAD$	$E\bar{B}DAF\bar{C}$
$E\bar{C}FAD\bar{B}$	$E\bar{C}\bar{B}DAF$
$FAE\bar{C}\bar{D}B$	$FAD\bar{C}EB$
$FE\bar{C}\bar{D}BA$	$FE\bar{B}A\bar{D}\bar{C}$
$F\bar{C}\bar{D}BAE$	$F\bar{C}EBAD$
$F\bar{D}BAE\bar{C}$	$F\bar{D}\bar{C}EBA$
$FBAE\bar{C}\bar{D}$	$FBDCEA,$

where, as before, the superscript line denotes a change of sign, $\bar{A} = -A$, &c.

As to the formation of this table, observe that we have $ABCDEF$ and $BCAF\bar{D}\bar{E}$; repeating the substitution, we have

$$\begin{aligned} &ABCDEF \\ &BCAF\bar{D}\bar{E} \\ &CABF\bar{E}\bar{D} \\ &\bar{A}BCDEF, \end{aligned}$$

viz. we have thus the $CABF\bar{E}\bar{D}$ which belongs to the third system; but nothing further, for the substitution is periodic of the third order, and the three forms are merely repeated. But if, upon

$$\begin{aligned} &ABCDEF \\ &BCAF\bar{D}\bar{E}, \end{aligned}$$

we operate successively with the substitutions which change the upper line into the three other forms of the first system

$$ADEFBC, \quad AEFBCD, \quad AFBCDE,$$

then the lower line is changed into the three forms

$$DEACF\bar{B}, \quad EFADB\bar{C}, \quad FBAED\bar{C},$$

which are forms belonging to the fourth, fifth, and sixth systems respectively. By way of verification, observe that, for instance repeating upon the second line the substitution

$$ABCDEF,$$

in place of

$$DEACF\bar{B},$$

we obtain

$$GFDABE,$$

which is one of the forms of the third system, assumed to have been previously found; and so in other instances.

Reverting to the three equations belonging to the form $ABCDEF$, by subtracting the third from the second equation, we obtain a linear relation between the four square roots C, D, E, F , viz. this is

$$0 = (\epsilon^2 - \epsilon^3)(C - F) + (\epsilon - \epsilon^4)(D - E);$$

and the same equation is obtained by means of the form $ABFEDC$. We thus obtain from the thirty lines of either column of the preceding table, thirty such equations, but obviously the number of such equations should be fifteen, for there can be but one relation between any four square roots C, D, E, F ; consequently each equation will be obtained twice; and, in fact, it is clear that the forms $ABCDEF$ and $BAFDEC$, and so in general any two forms which begin with the same pair of letters, give the same equation. But for greater symmetry, I write down the thirty equations in the order in which they are given by the left-hand column of the table: the equations are

$0 = (\epsilon^2 - \epsilon^3)$ multiplied by	$+ (\epsilon - \epsilon^4)$ multiplied by
$C - F$	$D - E$
$D - B$	$E - F$
$E - C$	$F - B$
$F - D$	$B - C$
$B - E$	$C - D$
$A + E$	$F + D$
$F - C$	$-D + E$
$-D - A$	$-E - C$
$-E - F$	$C - A$
$C + D$	$A - F$
$A + F$	$B + E$
$B - D$	$-E + F$
$-E - A$	$-F - D$
$-F - B$	$D - A$
$D + E$	$A - B$

(continuation of the table of fifteen equations from the left-hand column:)

$0 = (\epsilon^2 - \epsilon^3)$ multiplied by	$+ (\epsilon - \epsilon^4)$ multiplied by
$A + B$	$C + F$
$C - E$	$-F + B$
$-F - A$	$-B - E$
$-B - C$	$E - A$
$E + F$	$A - C$
$A + C$	$D + B$
$D - F$	$-B + C$
$-B - A$	$-C - F$
$-C - D$	$F - A$
$F + B$	$A - D$
$E - B$	$-C + D$
$-C - A$	$-D - B$
$-D - E$	$B - A$
$B + C$	$A - E$
$A + D$	$E + C$

where it is to be observed that, adding together the five equations given by any one of the systems, we obtain the identical result $0 = 0$.

I write down the 15 equations in a different order, in some cases changing the sign of the whole equation, as follows

$0 = (\epsilon^2 - \epsilon^3)$ multiplied by	$+ (\epsilon - \epsilon^4)$ multiplied by
$A + C$	$B + D$
$B + C$	$A - E$
$A + B$	$C + F$
$B + E$	$A - B$
$B + F$	$A - D$
$A + F$	$B + E$
$A + D$	$C + E$
$C + D$	$A - F$
$E + F$	$A - C$
$A + E$	$D + F$
$B - E$	$C - D$
$F - D$	$B - C$
$E - C$	$F - B$
$D - B$	$E - F$
$C - F$	$D - E$

The first three of these equations, or writing $\lambda = \epsilon^2 - \epsilon^3$, $\mu = \epsilon - \epsilon^4$, say

$$\begin{aligned}\lambda(A + C) + \mu(B + D) &= 0, \\ \lambda(B + C) + \mu(A - E) &= 0, \\ \lambda(A + B) + \mu(C + F) &= 0,\end{aligned}$$

constitute the entire system of independent linear relations between the square roots A, B, C, D, E, F . The coefficients λ, μ are such that

$$\lambda^2 - \mu^2 = \lambda\mu (= \epsilon - \epsilon^2 - \epsilon^3 + \epsilon^4, = \sqrt{5}),$$

and it is hence easy to verify that the remaining twelve equations can be deduced from these by the elimination of one or two of the square roots A, B, C . For instance, to eliminate A from the first and second equations, multiplying by $-\mu, \lambda$ and adding, we obtain

$$(-\lambda\mu + \lambda^2)C + (-\mu^2 + \lambda^2)B - \mu^2D - \lambda\mu E = 0,$$

that is,

$$\mu^2C + \lambda\mu B - \mu^2D - \lambda\mu E = 0,$$

or finally

$$\lambda(B - E) + \mu(C - D) = 0,$$

which is one of the equations. And so again eliminating A from the first and third equations, we find

$$\lambda(B - C) + \mu(C + F - B - D) = 0,$$

that is,

$$(\lambda - \mu)(B - C) + \mu(F - D) = 0,$$

or multiplying by λ ,

$$\mu^2(B - C) + \lambda\mu(F - D) = 0,$$

that is, finally

$$\lambda(F - D) + \mu(B - C) = 0,$$

which is one of the equations.

Cambridge, 21 March 1887.

868. On the Intersection of Curves.

[From the *Mathematische Annalen*, t. xxx. (1887), pp. 85–90.]

It is only recently that I have studied Bacharach's paper "Ueber den Cayley'schen Schnittpunktsatz," *Math. Ann.*, t. xxvi. (1885), pp. 275–299; his theorem in regard to the case where the δ points lie on a curve of the order $\gamma - 3$ is a very interesting and valuable one, but I consider it rather as an addition than as a correction to my original theorem; and I cannot by any means agree that the method by counting of constants is to be rejected as not trustworthy; on the contrary, it seems to me to be the proper foundation of the whole theory; it must of course be employed with due consideration of special cases. I reproduce the theorem in what appears to me the complete form.

Writing with Bacharach

$$r \rightleftharpoons m, \quad r \rightleftharpoons n, \quad r \rightleftharpoons m + n - 3, \quad \gamma = m + n - r, \quad \delta = \frac{1}{2}(\gamma - 1)(\gamma - 2),$$

and assuming $n \rightleftharpoons m$, (these conditions and inequalities are to be attended to throughout the present paper), I consider two curves of the orders m, n respectively meeting in δ points B , and in $(mn - \delta)$ points A ; and I state the theorem as follows:

1°. The $mn - \delta$ points A are in general such that a curve of the order r , which passes through $mn - \delta - 1$ of these points, does not of necessity pass through the remaining point; and in this case the general curve of the order r , which passes through the $mn - \delta$ points A , has for its form of equation

$$0 = \Omega_r + L_{r-m} P_m + M_{r-n} Q_n,$$

where $P_m = 0, Q_n = 0$ are the equations of the given curves, and L_{r-m}, M_{r-n} denote functions of the orders $r - m, r - n$ respectively; and it thus appears that the curve of the order r through the $mn - \delta$ points A passes also through the δ points B .

2°. If however the δ points B are on a curve of the order $\gamma - 3$, then the $mn - \delta$ points A are a system such that every curve of the order r passing through $mn - \delta - 1$ of these points passes through the remaining point; and in this case the general curve of the order r , which passes through the $mn - \delta$ points A , has for its form of equation

$$0 = \Omega_r + L_{r-m} P_m + M_{r-n} Q_n,$$

$\Omega_r = 0$ is a particular curve through the $mn - \delta$ points A , which does not go through any of the points B ; and consequently the curve of the order r does not pass through any of the points B .

For the proof of the theorem I premise as follows:

A curve of the order r depends upon $\frac{1}{2}r(r+3)$ constants, or, to use a convenient expression, its Postulandum is $= \frac{1}{2}r(r+3)$: if the curve has to pass through a given point, this imposes a single relation upon the constants, or say the Postulation is $= 1$; similarly, if the curve has to pass through k given points, this imposes k relations, or say the Postulation is $= k$. The points may be however a special system, for instance, they may be such that every curve of the order r which passes through them passes through one other point; the Postulation is in this case $= k - 1$; and so in other cases. Assuming the Postulation of the k points to be $= k$, then the Postulandum of a curve of the order r through the k points is $= \frac{1}{2}r(r+3) - k$. I stop to remark that the Postulation has reference to the particular curve or other entity in question; thus in the case of a curve passing through k points, the Postulation for a curve of a certain order may be $= k$, and for a curve of a different order it may be less than k :

Considering now, as above, two given curves of the orders m and n intersecting in the $mn - \delta$ points A and the δ points B , then assuming that the $mn - \delta$ points are not a special system, viz. that their Postulation in regard to a curve of the order r is $= mn - \delta$, the Postulandum of a curve of the order r through the $mn - \delta$ points is

$$\begin{aligned} &= \frac{1}{2}r(r+3) - mn + \delta, \\ &= \frac{1}{2}(r-m+1)(r-m+2) + \frac{1}{2}(r-n+1)(r-n+2) - 1, \end{aligned}$$

viz. this is identically true when for δ we write its value

$$= \frac{1}{2}(m+n-r-1)(m+n-r-2).$$

But we have through the $mn - \delta$ points A , a curve

$$L_{r-m} P_m + M_{r-n} Q_n = 0$$

with the proper Postulandum: viz. L_{r-m} contains $\frac{1}{2}(r-m+1)(r-m+2)$ constants, M_{r-n} contains $\frac{1}{2}(r-n+1)(r-n+2)$ constants, and there is a diminution -1 for the constant which divides out; hence this is the general equation of the curve of the order r through the $mn - \delta$ points A ; and the curve passes through the remaining δ points B .

In the case where the δ points B are on a curve of the order $\gamma - 3$, (observe that this is a single condition imposed on the δ , $= \frac{1}{2}(\gamma - 1)(\gamma - 2)$ points, for a curve of the order $(\gamma - 3)$ can be drawn through $\frac{1}{2}\gamma(\gamma - 3)$ points), it is to be shown that the Postulation of the $mn - \delta$ points A is $= mn - \delta - 1$; for, this being so, the Postulandum of the curve of the order r through the $(mn - \delta)$ points A is

$$= \frac{1}{2}r(r + 3) - mn + \delta + 1,$$

and the equation of the curve will be of the foregoing form, but it will be of the form

$$\Omega_r + L_{r-m}P_m + M_{r-n}Q_n = 0,$$

$\Omega_r = 0$ being a particular curve through the $mn - \delta$ points A , which does not pass through any of the points B . The proof depends on the theory of Residuation; which for the present purpose may be presented under the following form.

Let $A, B, \dots$ denote systems of points upon a given Basis-curve, for instance the foregoing curve $P_m = 0$, of the order m . And let $A \equiv ci$ denote that the points A are the complete intersection of the basis-curve by some other curve; (this implies that the number of the points is $= km$, a multiple of m , and the intersecting curve is then of course a curve of the order k). It is clear that, if $A \equiv ci$, and $B \equiv ci$, then also $A + B \equiv ci$. But conversely we have the theorem that, if $A + B \equiv ci$ and $A \equiv ci$, then also $B \equiv ci$. And we at once deduce the further theorem: if $A + B \equiv ci$, $B + C \equiv ci$, $C + D \equiv ci$, then also $A + D \equiv ci$. For the first and third relations give $A + B + C + D \equiv ci$, and the second relation then gives $A + D \equiv ci$.

[Diagram: a rectangle with vertices labelled A at top-left, B at top-right, D at bottom-left, C at bottom-right; the top side has length $mn - \delta$, divided as $mn - ab - \delta$ on the left portion and b on the right; the right side has length $\gamma - 3$; the bottom side $m(\gamma - 3) - b$; and the left side $m(\gamma - 3) + \delta$.]

Starting now (see the diagram) with the δ points B on a curve of the order $\gamma - 3$, suppose that we have through these points the basis-curve $P_m = 0$ of the order m , and another given curve $Q_n = 0$, of the order n ; and let these besides meet in the $mn - \delta$ points A . Let the curve of the order $\gamma - 3$ besides meet the basis-curve in the $m(\gamma - 3) - \delta$ points C ; and through these let there be drawn a curve of the order $m - 3$, which besides meets the basis-curve in the $m(r - n) + \delta$ points D . We have here $A + B \equiv ci$, $B + C \equiv ci$, $C + D \equiv ci$; consequently $A + D \equiv ci$, that is, the $mn - \delta$ points A and the $m(r - n) + \delta$ points D lie on a curve of the order r . The curve of the order $m - 3$ passes through the $m(\gamma - 3) - \delta$ points C ; its Postulandum is thus

$$= \frac{1}{2}m(m - 3) - m(\gamma - 3) + \delta,$$

which is

$$= \frac{1}{2} (r - n + 1)(r - n + 2).$$

In fact, substituting for γ, δ their values $= m + n - r$, and $\frac{1}{2} (m + n - r - 1)(m + n - r - 2)$ respectively, this equation is satisfied identically. The system of the $m(r - n) + \delta$ points D thus depends upon $\frac{1}{2} (r - n + 1)(r - n + 2)$ constants, or say the Postulandum of the points D is $= \frac{1}{2} (r - n + 1)(r - n + 2)$. It follows that the curve of the order r through the $(mn - \delta)$ points A and the $m(r - n) + \delta$ points D cannot have an equation of the form

$$L_{r-m} P_m + M_{r-n} Q_n = 0;$$

for the intersections of this curve with the basis-curve $P_m = 0$ are given by the equation $M_{r-n} Q_n = 0$, which contains only the

$$\frac{1}{2} (r - n + 1)(r - n + 2) - 1$$

constants of M_{r-n} (one constant of course divides out, giving the diminution -1). The equation must have the more general form

$$\Omega_r + L_{r-m} P_m + M_{r-n} Q_n = 0;$$

and it thus appears that the Postulation of the $mn - \delta$ points A , instead of being $= mn - \delta$, must be $= mn - \delta - 1$. This completes the proof.

I notice that, combining the last-mentioned identity

$$\frac{1}{2} m(m - 3) - m(\gamma - 3) + \delta = \frac{1}{2} (r - n + 1)(r - n + 2)$$

with the like identity

$$\frac{1}{2} n(n - 3) - n(\gamma - 3) + \delta = \frac{1}{2} (r - m + 1)(r - m + 2),$$

we obtain

$$\begin{aligned} & \frac{1}{2} m(m - 3) + \frac{1}{2} n(n - 3) - (m + n)(\gamma - 3) + 2\delta \\ &= \frac{1}{2} (r - n + 1)(r - n + 2) + \frac{1}{2} (r - m + 1)(r - m + 2), \end{aligned}$$

and consequently, referring to a former result, the left-hand side should be

$$= \frac{1}{2} r(r + 3) - mn + \delta + 1;$$

substituting for γ, δ their values, this is at once verified.

As appears by what precedes, Bacharach's special case is that in which the $\delta, = \frac{1}{2} (\gamma - 1)(\gamma - 2)$ points B satisfy the single condition of lying on a curve of the order $\gamma - 3$. We may have between the points B more than a single relation; in particular, the points B may be such as to include among themselves the complete intersection of two curves of the orders a, b respectively ($ab \leq \delta$): this will be the case, if the given curves are of the form

$$\begin{aligned} P_m &= \lambda_a S_{m-a} - \mu_b R_{m-b}, \\ Q_n &= \lambda_a V_{n-a} - \mu_b U_{n-b}, \end{aligned}$$

it being understood, here and in what follows, that the values of a, b are such that the suffixes are none of them negative.

The two curves here intersect in the ab points ($\lambda_a = 0, \mu_b = 0$), and in the $mn - ab$ points

$$\begin{vmatrix} \lambda_a & R_{m-b} & U_{n-b} \\ \mu_b & S_{m-a} & V_{n-a} \end{vmatrix} = 0,$$

say the $(mn - \delta)$ points A are $mn - ab - \Theta$ of the last-mentioned points, and the δ points B are the remaining Θ points together with the ab points. Here the general form of the curve of the order r passing through the $mn - ab$ points, and therefore through the $mn - \delta$ points A , is

$$\begin{vmatrix} L_{r-m-a+a+b} & M_{r-m} & N_{r-n} \\ \lambda_a & R_{m-b} & U_{n-b} \\ \mu_b & S_{m-a} & V_{n-a} \end{vmatrix} = 0,$$

where $L_{r-m-a+a+b}$, M_{r-m} , N_{r-n} are arbitrary functions of the orders indicated by the respective suffixes. The theory in regard to the number of constants is of course altogether different from that which belongs to the case of the general functions P_m , Q_n ; and it is probable that much interesting theory would present itself in the consideration of particular cases.

Cambridge, 22 March 1887.

869. On the Transformation of Elliptic Functions.

[From the *American Journal of Mathematics*, vol. IX. (1887), pp. 193–224.]

THE algebraical theory of the Transformation of Elliptic Functions was established by Jacobi in a remarkably simple and elegant form, but it has not hitherto been developed with much completeness or success. The cases $n = 3$ and $n = 5$ are worked out very completely in the *Fundamenta Nova* (1829); viz. considering the equation

$$\frac{M dy}{\sqrt{1 - y^2 \cdot 1 - \lambda^2 y^2}} = \frac{dx}{\sqrt{1 - x^2 \cdot 1 - k^2 x^2}},$$

($k = u^4$, $\lambda = v^4$; say this is the *Mkλ*- or *Mun*-form), Jacobi finds, in the two cases respectively, a modular equation between the fourth roots u , v , say the un-modular equation, and, as rational functions of u , v , the value of M and the values of the coefficients of the several powers of x in the numerator and denominator of the fraction which gives the value of y ; but there is no attempt at a like development of the general case. I shall have occasion to speak of other researches by Jacobi, Brioschi and myself; but I will first mention that my original idea in the present memoir was to develop the following mode of treatment of the theory:

In place of the *Mkλ*-form, using the $\rho q\Omega$ -form

$$\frac{dy}{\sqrt{1 - 2\beta y^2 + y^4}} = \frac{\rho dx}{\sqrt{1 - 2\alpha x^2 + x^4}}$$

(I write for greater convenience 2α , 2β in place of the α of Jacobi and Brioschi and the β of Brioschi), we can, by expanding each side in a series, integrating, and reverting the resulting series for y , obtain y in the form

$$y = \rho x (1 + \Pi_2 x^2 + \Pi_4 x^4 + \cdots).$$